DYNAMIC CPU FREQ: final report (raise only the minimum CPU clock)

Date: 2026-11-11. Board: VisionFive 2 (JH7110, 4x U74 @ up to 1.5 GHz). Compared with the **BASELINE** (`baseline_REPORT.md`): default CPU setting (ondemand, min 312.5 MHz), legacy vs hybrid, 1000 runs per mode.

The idea

Under the default `ondemand` setting the CPU sits at its lowest clock and a TLS handshake (a burst of a few ms) is over before the governor speeds up. Full speed (`performance`, fixed 1.5 GHz) fixes that but heats the chip. **Middle ground: keep `ondemand`, but raise only the minimum clock** (`scaling_min_freq`), so every handshake starts at a reasonable speed while the CPU can still save power.

Part 1: screening (`dynamic_cpu_freq_screen_REPORT.md`)

Four settings, 500 runs per mode each. On the LAN, ondemand + min 750 MHz (D1), min 1250 MHz (D2) and schedutil + min 750 MHz (D4) all cut the hybrid time by ~41 %; schedutil alone (D3) was worse than the default. **D1 was selected**: the lowest minimum with the full gain and the least heat.

Part 2: confirmation, D1 vs default, alternated block by block

Method: 20 blocks; in every block both CPU settings were measured back to back (order alternating), each with 50 legacy + 50 hybrid connections per scenario, so both settings saw the same network conditions. 1000 measurements per setting, mode and scenario; 8000 connections, 0 failures; every hybrid block verified (X25519MLKEM768 negotiated, ML-DSA-44 verified). Idle temperatures measured back to back in the same session. The CPU setting was restored to the default at the end. Files: `dynamic_cpu_freq_confirm_raw.csv`, `dynamic_cpu_freq_confirm_temps.csv`, `dynamic_cpu_freq_confirm.log`, `dynamic_cpu_freq_confirm_stats.txt`.

Setting		Governor	Minimum clock
D0	default (= baseline setting)	ondemand	312.5 MHz
D1	**dynamic CPU freq**	ondemand	**750 MHz**

Scenario 1: hackathon server (IPv4, 193.30.40.242), TLS handshake time

Setting	Mode	Best	Median	95th pct	Worst
D0 default	Legacy (X25519)	27.01 ms	**42.17 ms**	82.42 ms	407.89 ms
D0 default	Hybrid (X25519MLKEM768)	32.30 ms	**48.30 ms**	91.38 ms	873.22 ms
D1 min 750 MHz	Legacy (X25519)	26.10 ms	**37.16 ms**	68.83 ms	303.55 ms
D1 min 750 MHz	Hybrid (X25519MLKEM768)	29.25 ms	**40.26 ms**	75.03 ms	200.35 ms

Scenario 2: own server (LAN), end-to-end time (hybrid = ML-KEM + ECDSA P-256 + ML-DSA-44)

Setting	Mode	Best	Median	95th pct	Worst
D0 default	Legacy (X25519 + RSA-3072)	12.54 ms	**21.30 ms**	24.26 ms	30.59 ms
D0 default	Hybrid	20.12 ms	**27.84 ms**	31.55 ms	40.21 ms
D1 min 750 MHz	Legacy (X25519 + RSA-3072)	8.91 ms	**15.70 ms**	18.04 ms	21.74 ms
D1 min 750 MHz	Hybrid	13.93 ms	**16.19 ms**	19.22 ms	24.33 ms

Paired per-block comparison (same block = same network conditions)

Comparison	Hackathon server	Own server (LAN)
Hybrid: D1 vs D0	**−8.6 ms** median, D1 faster in 18/20 blocks	**−11.6 ms**, D1 faster in 20/20 blocks
Legacy: D1 vs D0	−6.5 ms, D1 faster in 17/20 blocks	−5.6 ms, D1 faster in 20/20 blocks
Hybrid − legacy at D0	+5.9 ms (hybrid faster in 0/20)	+6.4 ms (0/20)
Hybrid − legacy at D1	+4.5 ms (hybrid faster in 4/20)	**+0.5 ms** (4/20)

Temperature (°C)

Setting	Idle after 120 s	Under load, mean / max
D0 default	44.7	44.7 / 46.4
D1 min 750 MHz	45.2	44.6 / 46.3

For reference: full speed (`performance`, 1.5 GHz) idled at ~50–52 °C (2026-11-10).

Results against the target (baseline = legacy at the default clock)

Scenario	Baseline: legacy at default (D0)	Hybrid with dynamic CPU freq (D1)	Change
Hackathon server, handshake	42.17 ms	**40.26 ms**	**−1.9 ms (−4.5 %)** ✅
Own server, end-to-end (full hybrid auth)	21.30 ms	**16.19 ms**	**−5.1 ms (−24 %)** ✅

Findings

1. **The target is met with almost no heat cost.** With only the minimum clock raised to 750 MHz, hybrid PQC is faster than the legacy baseline at the default setting: −1.9 ms against the hackathon server and −5.1 ms for the full hybrid authentication on the LAN. The idle temperature rises by only +0.5 °C (45.2 vs 44.7 °C), versus about +6 °C for full speed.
2. **The hybrid penalty almost disappears on the device side.** At the default setting hybrid costs +6.4 ms on the LAN; at D1 only +0.5 ms. The remaining gap against the internet server (+3–4.5 ms) is mostly the larger ML-KEM messages on the network path.
3. **Reproducible.** The default setting re-measured in this run (D0) matches the baseline from earlier in the day (LAN: 21.30 vs 21.61 ms legacy, 27.84 vs 27.95 ms hybrid).
4. **Honest framing.** The CPU setting also speeds up legacy (−5.6 ms on the LAN). Under the same CPU setting hybrid is still slightly slower than legacy (+0.5 ms LAN, +3 to +4.5 ms internet). The claim is therefore: with a modest clock-policy change, hybrid PQC on this board is faster than the current (default) legacy deployment, at ~0.5 °C extra.
5. **Worst cases come from the network.** Single spikes of 200–870 ms against the internet server appear in both modes and both settings; on the LAN the worst case is 24 ms (D1 hybrid).

RECOMMENDATION: which CPU clock setting to use

Optimal setting (of those tested): `ondemand` governor with the minimum clock raised to 750 MHz (D1). It's the lowest minimum that gives the full speed-up, it's the coolest of the settings that work, and it keeps the power-saving governor (the CPU still goes up to 1.5 GHz when needed and stays at 750 MHz when idle).

All settings compared: own server (LAN), full hybrid authentication, end-to-end median

Setting	Governor / clock	Legacy	Hybrid	Hybrid – legacy	Idle temp	Runs per mode	Verdict
Default (baseline)	ondemand, min 312.5 MHz	21.30 ms	27.84 ms	+6.5 ms	44.7 °C	1000	❌ hybrid slower than baseline
D1	**ondemand, min 750 MHz**	**15.70 ms**	**16.19 ms**	**+0.5 ms**	**45.2 °C**	**1000**	✅ **recommended**
D2	ondemand, min 1250 MHz	17.54 ms	16.67 ms	−0.9 ms	46.6 °C	500	✅ same speed, warmer
D3	schedutil, min 312.5 MHz	23.90 ms	33.85 ms	+10.0 ms	46.2 °C	500	❌ worse than default
D4	schedutil, min 750 MHz	17.58 ms	16.54 ms	−1.0 ms	46.6 °C	500	✅ same speed, warmer
Full speed	performance, fixed 1.5 GHz	10.66 ms	10.32 ms	−0.3 ms	~50–52 °C	100	✅ fastest, hottest

Notes on comparability: - Default and D1 come from the same alternating 1000-run confirmation, so they're directly comparable. - D2–D4 come from the screening run (sequential, 500 runs); their LAN numbers are reliable, and their temperatures were taken the same morning. - Full speed was measured on 2026-11-10 (50 runs x 2 passes, different day and room temperature), so treat its numbers as indicative. - Against the hackathon server (internet), only the default vs D1 comparison is clean: hybrid 48.30 → 40.26 ms, below the legacy baseline of 42.17 ms.

Why D1 and not the others

Option	Why not chosen
Default (min 312.5 MHz)	Handshakes start at the lowest clock, so hybrid is +6.5 ms slower than legacy and misses the target.
D2 (min 1250 MHz), D4 (schedutil + 750 MHz)	Same latency as D1 (within ~0.5 ms) but about 1 °C warmer at idle. No benefit.
D3 (schedutil alone)	Reacts too slowly for handshakes lasting a few ms; worse than the default.
Full speed (1.5 GHz fixed)	About 6 ms faster again on the LAN, but ~5–7 °C hotter at idle; the user switched it off because of heat. Use it only if a heatsink/fan is fitted and latency matters most.

Choosing by priority

Priority	Use
Lowest latency, heat not a problem (heatsink/fan)	`performance` (fixed 1.5 GHz)
Meet the latency target with minimal heat (recommended)	**`ondemand` + min 750 MHz**
Lowest heat/energy, latency not critical	default `ondemand` (min 312.5 MHz)

Not yet tested

- Minimum clocks of 500 MHz and 625 MHz: a lower minimum might give the same gain with even less heat. 750 MHz is optimal among the tested settings, not proven to be the absolute optimum.

- Energy use (needs a USB power meter). Temperature was used as the proxy.

How to apply D1 on the board

```
echo 750000 | sudo tee /sys/devices/system/cpu/cpufreq/policy0/scaling_min_freq     # now
(until reboot)
echo 312500 | sudo tee /sys/devices/system/cpu/cpufreq/policy0/scaling_min_freq     # back to
default
```

Current state: **default** (min 312.5 MHz). D1 isn't made permanent.

Source: dynamic_cpu_freq_REPORT.md